

ECON 5345 Homework 3 Report

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Question 1

Question 2

a. Note that we can write

$$\text{Var}(e) = [\Gamma_{ij}]_{i,j=1}^n \in \mathbb{R}^{Tn \times Tn},$$

where $\Gamma_{ij} = \text{Cov}(e_i, e_j) \in \mathbb{R}^{T \times T}$ is the (i, j) -th block of $\text{Var}(e)$. Since v_t is white noise, we have

$$\text{Cov}(e_{it}, e_{j\tau}) = \begin{cases} \Omega_{ij} \in \mathbb{R}, & t = \tau \\ 0, & \text{otherwise} \end{cases}$$

Therefore, we have $\Gamma_{ij} = \text{diag}(\Omega_{ij}) = \Omega_{ij} I_T$. Hence,

$$\text{Var}(e) = \Omega \otimes I_T.$$

b. Since Ω is a covariance matrix, it is positive definite. Hence, Ω is non-singular. By the properties of Kronecker product, we have

$$\begin{aligned} \hat{\beta}_{\text{GLS}} &= ((I_n \otimes X')(\Omega^{-1} \otimes I_T)(I_n \otimes X))^{-1} ((I_n \otimes X')(\Omega^{-1} \otimes I_T)y) \\ &= (\Omega^{-1} \otimes X'X)^{-1} (\Omega^{-1} \otimes X')y \\ &= (\Omega \otimes (X'X)^{-1}) (\Omega^{-1} \otimes X')y \quad \text{since } X'X \text{ is positive-definite} \\ &= (I_n \otimes (X'X)^{-1} X') y \\ &= (I_n \otimes (X'X)^{-1}) (I_n \otimes X')y \\ &= (I_n \otimes (X'X)^{-1}) ((I_n \otimes X)'y) \\ &= ((I_n \otimes X)'(I_n \otimes X))^{-1} ((I_n \otimes X)'y) \\ &= \hat{\beta}_{\text{OLS}}. \end{aligned}$$

Question 3

a. 1